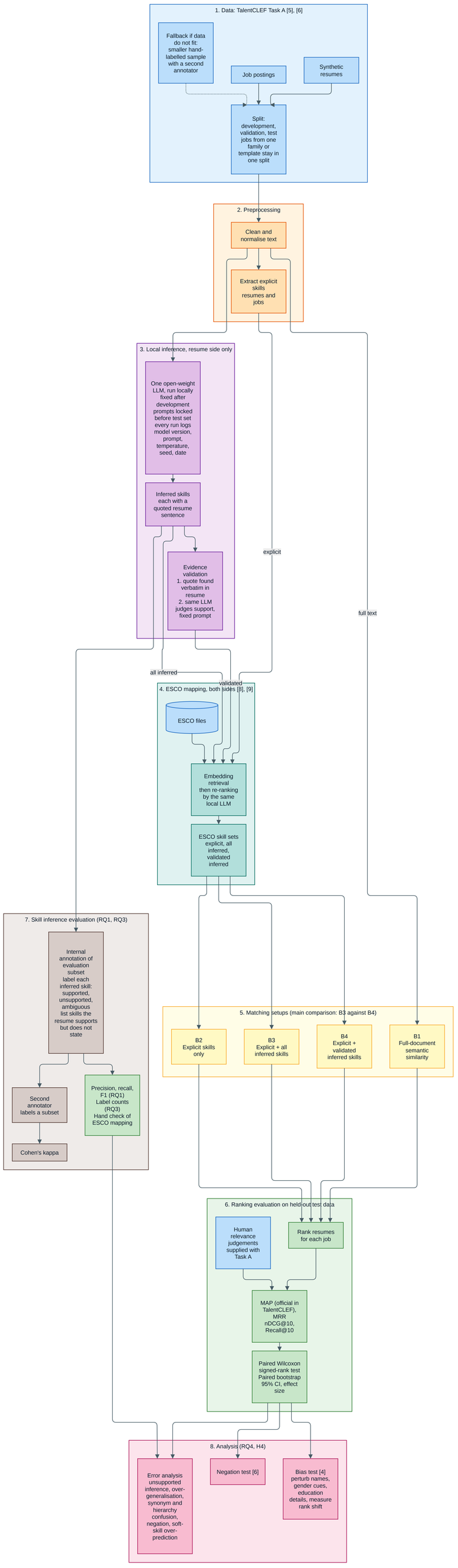
1. Data: TalentCLEF Task A [5], [6]
Fallback if data do not fit: smaller hand-labelled sample with a second annotator
Job postings
Synthetic resumes
Split: development, validation, test jobs from one family or template stay in one split
2. Preprocessing
Clean and normalise text
Extract explicit skills resumes and jobs
3. Local inference, resume side only
One open-weight LLM, run locally fixed after development prompts locked before test set every run logs model version, prompt, temperature, seed, date
Inferred skills each with a quoted resume sentence
Evidence validation
1. quote found verbatim in resume
2. same LLM judges support, fixed prompt
explicit
full text
all inferred
validated
4. ESCO mapping, both sides [8], [9]
ESCO files
Embedding retrieval then re-ranking by the same local LLM
ESCO skill sets explicit, all inferred, validated inferred
7. Skill inference evaluation (RQ1, RQ3)
Internal annotation of evaluation subset label each inferred skill: supported, unsupported, ambiguous list skills the resume supports but does not state
Second annotator labels a subset
Cohen's kappa
Precision, recall, F1 (RQ1) Label counts (RQ3) Hand check of ESCO mapping
5. Matching setups (main comparison: B3 against B4)
B2 Explicit skills only
B3 Explicit + all inferred skills
B4 Explicit + validated inferred skills
B1 Full-document semantic similarity
6. Ranking evaluation on held-out test data
Human relevance judgements supplied with Task A
Rank resumes for each job
MAP (official in TalentCLEF), MRR nDCG@10, Recall@10
Paired Wilcoxon signed-rank test Paired bootstrap 95% CI, effect size
8. Analysis (RQ4, H4)
Error analysis unsupported inference, over-generalisation, synonym and hierarchy confusion, negation, soft-skill over-prediction
Negation test [6]
Bias test [4] perturb names, gender cues, education details, measure rank shift